SILICAFLUX

FROM SiO_2 TO FOUNDRY

A practical guide to the materials, processes, equipment and infrastructure needed to manufacture a custom silicon semiconductor

PURPOSE

This report maps the complete pathway from high-purity silica (SiO_2) to a packaged SilicaFlux integrated circuit. It explains what must be produced, what must be purchased or qualified from suppliers, what a foundry actually does, and how to stage development without prematurely committing to a full wafer-fabrication plant.

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Status: Conceptual engineering roadmap - not a process recipe, plant design, or investment-grade feasibility study.

Executive summary

SilicaFlux needs to master a chain of distinct disciplines. High-purity silica is a useful starting mineral, but it is not itself a semiconductor and it is not sufficient on its own to make a chip. The silica is first converted to metallurgical silicon, then purified to electronic-grade polysilicon, grown into single-crystal ingots and processed into wafers. A foundry then builds transistors and interconnects on those wafers using a tightly controlled sequence of oxidation, deposition, lithography, etching, doping, annealing, cleaning, planarisation and inspection.

The finished chip also depends on materials that do not come from silica: dopants, conductive metals, dielectric materials, photoresists, process gases, wet chemicals, photomasks, specialty ceramics, packaging substrates and test hardware. The production facility requires cleanrooms, ultra-pure water, high-purity gas and chemical distribution, vacuum systems, exhaust abatement, wastewater treatment, reliable power and extensive metrology.

THE MOST IMPORTANT STRATEGIC POINT

Do not build a complete foundry as the first milestone. Design the SilicaFlux processor against a qualified foundry PDK and manufacture the first ASIC through an established foundry. In parallel, prove the silica-to-silicon materials route and develop proprietary capabilities where they create a measurable cost, quality or supply advantage.

Key recommendations

- Separate the mineral-to-wafer business from the wafer-to-chip foundry business. They have different process technologies, quality systems, capital requirements and customer bases.
- Treat electronic-grade silicon purity as an elemental impurity specification, not a single headline SiO_2 percentage or a generic “11N” claim.
- Choose the first fabrication process from a real foundry PDK, design rules, device models, standard-cell library, SRAM options and production support - not from a node-size slogan.
- Keep wafer fabrication outsourced for the first chip. A small design team can validate architecture and silicon before the company considers owning process equipment.
- Invest early in process safety, contamination control, metrology, traceability and yield learning. These are core manufacturing capabilities, not afterthoughts.

The complete chain at a glance

Upstream materials	Wafer manufacturing	Chip fabrication and delivery
Quartz mining and assay	Single-crystal growth	Oxidation / deposition
Beneficiation and purification	Ingot machining and wafering	Lithography / etch / doping
Carbothermic reduction	Wafer polish and qualification	Metallisation / CMP / inspection
Chlorosilane purification	Wafer supply and traceability	Wafer test / dicing / packaging

1. Start with the material distinction

The terms silica, silicon and silicon dioxide are related but not interchangeable. Keeping them distinct prevents major errors in process design and commercial planning.

Term	Meaning	Role in SilicaFlux
Silica / silicon dioxide (SiO_2)	A compound of silicon and oxygen. Quartz is a crystalline mineral form of SiO_2.	Mineral feedstock; also a source for fused quartz and oxide materials.
Metallurgical-grade silicon (MG-Si)	Elemental silicon produced by reducing silica, typically with carbon in an electric furnace.	Intermediate metal for further purification.
Electronic-grade polysilicon	Very high-purity elemental silicon produced through chemical purification and deposition.	Feedstock for semiconductor single-crystal growth.
Monocrystalline silicon	A single-crystal silicon ingot with controlled orientation and electrical properties.	Material from which semiconductor wafers are sliced.
Silicon wafer	A thin, polished, qualified slice of single-crystal silicon.	Substrate entering a CMOS foundry.

IMPORTANT

A wafer can be made from silicon that was originally extracted from SiO_2, but the active transistor is made in engineered elemental silicon. SiO_2 commonly serves as an insulator or process material in the device; it is not the conducting channel in a conventional silicon MOSFET.

The upstream route: SiO_2 to wafer

1. Mine and characterise quartz or another suitable silica source; analyse mineral inclusions and trace elements.
2. Beneficiate the material through selective crushing, sizing, sorting, physical separation and appropriately controlled chemical cleaning.
3. Reduce SiO_2 to metallurgical silicon in a submerged-arc furnace using a controlled carbon reductant system.
4. Convert metallurgical silicon into a volatile silicon compound, commonly trichlorosilane, and purify it by distillation.
5. Deposit high-purity polysilicon through a chemical vapour deposition process (the Siemens process is the established industrial route).
6. Grow a single-crystal ingot, typically by the Czochralski method for mainstream silicon wafers; machine, slice, polish, clean and qualify it.

OSHA's semiconductor substrate-manufacturing overview describes the electric-furnace reduction, trichlorosilane purification, polysilicon deposition, Czochralski growth and wafer-preparation stages [1]. The actual plant design and impurity limits must be specified with experienced process engineers and customers.

2. Materials and quality gates before the foundry

2.1 Quartz feedstock qualification

Do not select a deposit only because a laboratory certificate reports a high bulk SiO_2 percentage. Semiconductor suitability depends on the identity, concentration, location and removability of individual impurities. Surface contamination, mineral inclusions and lattice-bound impurities behave differently during beneficiation and reduction.

Control area	What to measure	Why it matters
Bulk chemistry	SiO_2 and major oxides	Establishes the overall feedstock composition.
Trace elements	Al, Fe, Ti, Ca, Na, K, Li, B, P and other relevant elements	Predicts purification burden and possible transfer to silicon.

Control area	What to measure	Why it matters
Mineralogy and inclusions	X-ray diffraction, microscopy, inclusion analysis	Identifies feldspar, mica, iron minerals and trapped phases.
Physical properties	Particle size, liberation, moisture, hardness	Influences beneficiation, furnace behaviour and handling.
Consistency	Variability between faces, benches and production lots	Determines whether the deposit can support repeatable manufacturing.

2.2 Silicon and wafer quality controls

Quality must be assessed at every transition. A strong upstream assay does not prove the final polysilicon, crystal or wafer meets a customer specification. The qualification programme should include appropriate methods such as GDMS or ICP-MS for trace elements, FTIR for interstitial oxygen and carbon in silicon, SIMS for selected dopants and contaminants, X-ray methods for crystal orientation, and surface/particle inspection for wafers. The laboratory should use validated methods, reference materials, controlled sampling and a documented chain of custody.

QUALITY GATE

Write the product specification backwards from the intended wafer customer and foundry process. Agree the required resistivity, dopant type, crystal orientation, oxygen/carbon range, defect density, surface finish, particle levels and packaging conditions before buying major equipment.

3. What the foundry needs beyond silicon and SiO_2

A CMOS process uses multiple classes of materials. The exact bill of materials depends on the selected process technology and must come from the foundry process integration team and its PDK. The list below is a technology map, not a universal recipe.

Material family	Examples	Typical function
Semiconductor substrate	Monocrystalline silicon; specialist SiGe in selected processes	Active device body or engineered channel structures.
Insulators and dielectrics	Thermal SiO_2, deposited SiO_2, silicon nitride, high-k dielectrics such as HfO_2	Gate insulation, isolation, spacers, passivation and interlayer dielectric.
Dopant sources	Boron, phosphorus, arsenic, antimony compounds or ion sources	Create p-type and n-type regions in controlled locations.
Conductors and barriers	Copper, aluminium, tungsten, titanium, tantalum, nitrides and related stacks	Contacts, local wiring, interconnects and diffusion barriers.
Lithography materials	Photoresist, developers, anti-reflective coatings, solvents, photomasks	Transfer designed patterns to wafer layers.
Process gases and precursors	Nitrogen, argon, oxygen, hydrogen and process-specific deposition/etch gases	Purging, oxidation, deposition, plasma processing and chamber control.
Wet-process chemicals	High-purity acids, bases, oxidisers, solvents and rinse water	Wafer cleaning, surface preparation, etching and residue removal.
Consumables and components	CMP slurries/pads, filters, quartzware, ceramics, seals and wafer carriers	Support repeatable processing and contamination control.

3.1 SiO_2 has several roles

- Thermal oxide: silicon at the wafer surface is deliberately oxidised to form an insulating film.
- Deposited oxide: SiO_2 can be deposited by process-specific CVD or other film methods for isolation and interlayer insulation.
- Fused quartz components: high-purity silica is used in selected furnace and process components, subject to compatibility and contamination specifications.
- Polishing materials: colloidal silica can be used in certain chemical-mechanical polishing slurries; slurry formulation and particle control are critical.

3.2 Why SiO_2 is not the only dielectric

Very thin SiO_2 layers become increasingly susceptible to direct tunnelling leakage as device dimensions shrink. Some advanced transistor processes therefore use high-k dielectrics, often combined with metal gates. Other processes use SiO_2, silicon nitride or layered dielectric stacks for different functions. SilicaFlux should not assume that the highest-purity natural SiO_2 can replace every dielectric material in a modern foundry.

4. How a silicon foundry builds the circuit

Front-end fabrication is a repeated sequence of adding, patterning, modifying and measuring materials on a wafer. The same wafer may pass through many hundreds of process steps, with inspection and cleaning interleaved throughout. The exact order and count depend on the selected process integration scheme [2].

Module	What happens	Key equipment / control
Wafer clean and preparation	Remove particles, organics, metals and native surface films as required.	Wet benches, single-wafer cleaners, megasonic systems, ultrapure water and chemical monitoring.
Oxidation	Grow silicon oxide on exposed silicon where required.	Batch or single-wafer oxidation furnaces; gas flow, temperature and film-thickness control.

Module	What happens	Key equipment / control
Thin-film deposition	Add dielectric, semiconductor or conductive films.	CVD, PVD, ALD and epitaxy tools; precursor purity and uniformity control.
Photolithography	Coat resist, align a mask, expose and develop a pattern.	Track, coater/developer, stepper or scanner, reticle handling and overlay metrology.
Etch	Remove selected exposed materials to create patterned features.	Wet etch benches and plasma etchers; endpoint and selectivity control.
Doping and anneal	Introduce dopants and activate them while controlling diffusion.	Ion implanters, diffusion furnaces and rapid thermal processing.
Metallisation	Create contacts and metal interconnect layers.	PVD/CVD/ALD, electroplating where applicable, barrier/liner processes and etch.
CMP and planarisation	Flatten the wafer before subsequent layers.	CMP tools, slurry delivery, pads, post-CMP clean and thickness metrology.
Metrology and inspection	Measure films, critical dimensions, defects, overlay and electrical properties.	Ellipsometry, reflectometry, CD-SEM, optical inspection, profilometry and electrical test.

PROCESS INTEGRATION

A foundry is not a set of independent machine purchases. Each process step changes the surface and thermal history for later steps. Process integration engineers must define the full sequence, interfaces, control windows and acceptance limits for the chosen transistor architecture.

5. Foundry equipment and factory systems

The table below separates wafer-process tools from the infrastructure needed to keep them operating. A real equipment list must be built from a selected process flow, required wafer diameter, capacity target, uptime model and vendor support plan.

Area	Equipment examples	What to specify
Wafer handling	FOUPs/cassettes, robots, wafer aligners, stockers	Wafer diameter, automation level, particle performance and interface compatibility.
Lithography	Resist coat/develop tracks, aligners, steppers/scanners, mask handling	Minimum feature capability, overlay, throughput, resist compatibility and service support.
Deposition	Oxidation furnaces, LPCVD/PECVD, PVD, ALD, epitaxy	Film materials, thickness range, uniformity, contamination and chamber matching.
Etch and clean	Wet benches, plasma etchers, strip tools, single-wafer cleaners	Material selectivity, residues, damage control and chemical compatibility.
Doping and thermal	Ion implanters, diffusion furnaces, RTP tools	Dose/energy range, uniformity, activation, thermal budget and contamination control.
CMP	Polish tools, slurry distribution, post-CMP cleaners	Planarity, dishing/erosion, defects, endpoint and consumables cost.
Metrology	Film and CD metrology, defect inspection, overlay and electrical test	Measurement uncertainty, sampling strategy, calibration and data integration.
Back-end test	Wafer probers, dicing saws/lasers, die bond, wire bond or flip-chip tools	Test coverage, package style, thermal/electrical requirements and throughput.

5.1 Factory utilities and supporting systems

System	Why a fab needs it
Cleanroom HVAC and filtration	Controls airborne particles, temperature, humidity, pressure cascade and heat loads.
Ultra-pure water (UPW)	Provides highly controlled rinse and process water; water quality and distribution require continuous monitoring.
Bulk and specialty gas systems	Store, purify, distribute and monitor inert, reactive and process gases safely.
Chemical distribution	Delivers high-purity wet chemicals in compatible piping, tanks and point-of-use systems.
Vacuum and exhaust	Supports process chambers and removes process gases through designed collection systems.
Abatement and scrubbers	Treat hazardous or environmentally regulated exhaust streams before release.
Wastewater treatment	Segregates and treats acidic, alkaline, solvent-bearing and metal-bearing waste streams.
Power and thermal management	Supports high, stable loads; protects tools from interruptions and manages chilled water/heat rejection.
Facilities monitoring and automation	Tracks alarms, environmental status, tool state, chemical inventory and maintenance.

SEMI publishes guidance on ultrapure-water system design and filter performance for semiconductor processing [3]. A fab must define UPW quality and monitoring with its process requirements rather than assume ordinary deionised water is sufficient.

6. Cleanroom, contamination control and process safety

6.1 Contamination control

A cleanroom is a controlled production environment, not simply a room with filters. The factory needs particle control, controlled airflow, pressure cascades, personnel gowning, material transfer rules, equipment exhaust, vibration management and a contamination-control strategy. Cleanroom classification should be chosen for each area based on the process and risk, using the applicable ISO 14644 framework and customer/foundry requirements.

- Use qualified wafer containers and controlled material-handling routes.
- Separate wet chemistry, lithography, high-temperature processes and metal contamination-sensitive areas where the process requires it.
- Establish particle, airborne molecular contamination and metallic contamination monitoring appropriate to the process.
- Control cross-contamination through dedicated tools, chamber seasoning, maintenance procedures and documented recovery steps.
- Use statistically defined sampling and excursion management so defects are detected before value is added by later process steps.

6.2 Process safety and environmental management

A semiconductor fab handles crystalline silica dust in upstream material operations, high-temperature furnaces, corrosive and oxidising chemicals, flammable and pyrophoric gases, vacuum systems, electrical hazards and potentially toxic exhaust products. The plant must have a formal process-hazard analysis and engineered safeguards designed by competent specialists.

Risk area	Required management approach
Crystalline silica dust	Enclosed transfer, local exhaust ventilation, exposure monitoring, housekeeping and occupational-health controls.
Reactive / toxic gases	Gas cabinets, automatic shutoff, leak detection, ventilation, interlocks, emergency response and validated abatement.
Wet chemicals	Compatible storage and delivery, secondary containment, segregated drainage, detection and emergency showers/eyewash.
High temperature / furnaces	Thermal guarding, interlocks, maintenance isolation, cooling-water monitoring and emergency procedures.
Waste and emissions	Characterise streams, segregate incompatible wastes, treat exhaust and wastewater, and comply with permits.
Operational continuity	Preventive maintenance, spare parts, backup power for critical systems, disaster recovery and business continuity.

OSHA’s semiconductor manufacturing guidance lists process hazards and controls for silicon substrate production, device fabrication, metallisation and assembly [1,2]. The specific legal framework for a UK facility would also include UK health-and-safety, environmental permitting, dangerous-substances and waste requirements; these must be assessed by qualified UK advisers before design or procurement.

7. The design-to-foundry bridge: EDA, PDK and tape-out

SilicaFlux's processor architecture becomes a physical chip through an electronic-design-automation (EDA) flow. The process design kit (PDK) supplied by the selected foundry is the interface between the logical design and the manufacturing process. It defines process-specific design rules, device models, standard cells, layout layers and other implementation data.

7.1 Required design flow

Step	Deliverable / activity	Why it matters
Architecture specification	Instruction set, compute units, memory hierarchy, bus/interconnect, performance and power targets.	Defines what the chip must do and how it is partitioned.
RTL implementation	Synthesizable SystemVerilog or another supported hardware description.	Describes the digital logic to be implemented.
Functional verification	Simulation, assertions, formal checks and testbenches.	Finds functional errors before expensive fabrication.
Synthesis	Maps RTL into the PDK's standard-cell libraries.	Creates a gate-level implementation supported by the foundry process.
Physical design	Floorplan, power delivery, placement, clock tree and routing.	Turns logic into geometric layout with electrical and timing constraints.
Sign-off checks	DRC, LVS, timing, power integrity, antenna and other foundry-required checks.	Verifies that the layout conforms to process and electrical constraints.
Tape-out	Release of foundry-accepted layout data and manufacturing documentation.	Starts wafer fabrication.
Bring-up and validation	Wafer probe, package test, debug, performance and reliability characterisation.	Determines whether manufactured silicon meets the specification.

7.2 What to look for in a foundry PDK

- A production-qualified process and clearly defined supported design rules.
- Transistor models, standard-cell libraries, I/O libraries and SRAM or memory compiler options where needed.
- Supported EDA tools and a documented digital implementation/sign-off flow.
- Multi-project wafer or prototyping access, engineering support and a clear shuttle schedule.
- Commercial terms covering confidentiality, data handling, IP ownership, yield responsibility and production availability.

> **CAUTION ON OPEN PDKS**
>
> The public SkyWater SKY130 PDK is useful for learning and early design exploration, but its documentation labels the release as an experimental preview and says it is not intended for production settings. Treat it as a prototype resource; confirm current foundry status and production authorization before relying on it for commercial tape-out [4].

8. SilicaFlux chip architecture and packaging

8.1 First-chip scope

The existing SilicaFlux concept includes scalar, vector and tensor processing, a dataflow scheduler, local SRAM and an on-chip interconnect. For a first physical chip, prioritise a compact demonstrator with a clear test plan rather than a feature-complete commercial processor.

Block	First-silicon design objective	Key engineering risk
Scalar control core	Run basic control, configuration and test firmware.	Boot flow, interrupts, debug and verification completeness.

Block	First-silicon design objective	Key engineering risk
SIMD/vector engine	Demonstrate parallel arithmetic on small, repeatable workloads.	Datapath correctness, timing closure and data delivery.
Tensor unit	Demonstrate a small matrix operation with reference test vectors.	Numerical behaviour, throughput and SRAM bandwidth.
SRAM and memory interface	Provide local data storage and external access.	Macro availability, timing, power and bandwidth bottlenecks.
Interconnect / scheduler	Move work and data between the compute blocks.	Deadlock, arbitration, congestion and observability.
Test and debug logic	Enable wafer test, scan, basic debug and failure isolation.	Test coverage and diagnosis after fabrication.

8.2 Package and test

After wafer fabrication, dies are electrically screened, separated, attached to a package or substrate, electrically connected, protected and tested. The packaging choice affects thermal resistance, signal integrity, power delivery, cost and manufacturability. A simple package is adequate for an initial demonstrator; advanced packaging and chiplets should follow only when there is a demonstrated bandwidth, density or product need.

1. Wafer probe: test dies before dicing and identify known-good candidates.
2. Dicing: separate individual dies using a qualified cutting process.
3. Die attach and interconnect: attach the die and connect it through wire bond, flip-chip or another package-specific approach.
4. Encapsulation and lid / thermal interface: protect the die and provide a controlled thermal path.
5. Final test: verify function, speed bins, power, temperature range and reliability requirements.

9. What SilicaFlux should own - and what it should buy

Vertical integration is a long-term option, not an automatic first step. Each step should be internalised only when it offers a defensible improvement in supply security, cost, performance, intellectual property or customer qualification.

Capability	Recommended initial model	Potential in-house path
Quartz mining / feedstock	Secure qualified supply; run independent assay and sampling.	Own or partner in a deposit after reserve, permitting and beneficiation tests.
Beneficiation	Pilot plant or specialist mineral-processing partner.	Bring in-house when feedstock quality and scale are proven.
Metallurgical silicon	Contract producer or toll-processing partner for early trials.	Build furnace capacity only after impurity and energy economics are validated.
Electronic-grade polysilicon	Purchase qualified product or collaborate with an established producer.	Consider only with major capital, chemical-process expertise and offtake.
Crystal growth and wafers	Buy qualified wafers for first chip; use specialist wafer supplier.	Develop specialist wafer capability if quality, supply or IP case is strong.
Chip design	Own SilicaFlux architecture, RTL, verification and software.	Core proprietary capability from the start.
Wafer fabrication	Use foundry services and a production-qualified PDK.	Consider mature-node fab only after stable demand and process expertise.
Packaging and test	Use OSAT / specialist assembly and test partners.	Bring test engineering and selected packaging capability in-house over time.

9.1 Why not build the complete foundry first?

A wafer fab requires a coordinated process platform, trained process and equipment engineers, qualified materials, production data, yield-learning systems, spare parts, vendor service contracts, a cleanroom utility backbone and enough customer demand to keep tools economically utilised. Owning isolated tools without a qualified integrated process does not create a working foundry.

The UK National Semiconductor Strategy (2023) stated that a state-of-the-art advanced silicon foundry could cost more than £18 billion. This is a policy-era estimate for state-of-the-art facilities, not a price quote for a mature-node pilot line, a small demonstrator or any particular SilicaFlux configuration [5].

10. Recommended SilicaFlux development roadmap

Phase	Main work	Exit gate / evidence required
0. Define product	Set chip use case, workload, volume, performance, power, package and target market.	A product specification and business case with measurable requirements.
1. Prove architecture	Resolve RTL/toolchain, verify scalar/vector/tensor blocks, prototype on FPGA.	Repeatable benchmark results, test coverage and stable RTL.
2. Select foundry	Compare mature-node processes, PDK, IP availability, wafer options, support and commercial terms.	Foundry process selected; PDK and design flow validated.
3. First tape-out	Complete synthesis, physical design, sign-off, tape-out and package/test planning.	Manufacturable layout accepted by foundry.
4. First silicon	Probe, package, bring up, debug and characterise the chip.	Working silicon, measured power/performance and failure analysis.
5. Materials pilot	Test quartz beneficiation, reduction and purification routes with mass balance and impurity tracking.	Demonstrated repeatable silicon quality and credible cost/energy data.

Phase	Main work	Exit gate / evidence required
6. Wafer / oxide R&D	Evaluate crystal growth, wafer qualification, fused quartz and SiO_2 film capabilities.	Customer-relevant material specifications and process capability.
7. Scale decision	Compare buy/partner/build options for each upstream and fab stage.	Investment decision supported by offtake, yield, utilities, permits and capital model.

10.1 Node selection: choose by process capability, not headline nanometres

For an initial ASIC, a mature process node is generally more accessible, lower risk and better supported for simple demonstrator designs than a leading-edge process. However, “130 nm”, “180 nm”, “65 nm” and “28 nm” are not interchangeable product specifications. Select a node only after reviewing the actual foundry PDK, available logic and memory IP, voltage options, metal stack, design rules, process maturity, shuttle availability, expected die area and total cost.

11. Manufacturing metrics, controls and decision gates

SilicaFlux should manage the materials chain and chip programme through linked quality, yield, cost and reliability metrics. Every stage needs a named owner, a control plan, a sampling method, a non-conformance procedure and a traceable lot record.

Area	Core metrics
Mineral feedstock	SiO_2 assay; trace impurity profile; inclusion count/type; beneficiation recovery; lot-to-lot variance.
Reduction	Silicon recovery; furnace energy per kg; reductant consumption; impurity transfer; off-gas and slag balance.
Polysilicon	Elemental impurity results; deposition yield; batch consistency; chemical recovery and recycle; customer acceptance.
Crystal / wafer	Crystal yield; resistivity uniformity; oxygen/carbon; dislocation/defect density; wafer flatness, thickness and surface particles.
Foundry process	Tool availability; process capability; defect density; wafer yield; critical-dimension and overlay control; excursion rate.
Chip product	Functional yield; performance; power; thermal behaviour; package reliability; test coverage; field returns.
Economics	Cost per accepted wafer/die; energy per accepted kg or wafer; scrap; consumables; maintenance; total conversion cost.

11.1 Data and traceability architecture

Build a batch-level digital record linking mine block, quartz lot, beneficiation run, reductant lot, furnace campaign, chemical purification batch, polysilicon batch, crystal/ingot, wafer lot, foundry lot and packaged chip. Where confidentiality permits, connect material genealogy to electrical performance and wafer yield. This data asset can help isolate contamination sources and improve process windows.

12. Final priorities for SilicaFlux

PRIORITY 1 - GET A CHIP INTO SILICON

Own the processor architecture and verification. Select a foundry with an appropriate mature-node process and a production-capable PDK. Manufacture and test a small ASIC before committing to a wafer fab.

PRIORITY 2 - PROVE THE MATERIAL ADVANTAGE

Run controlled trials from selected quartz through beneficiation and silicon reduction. Determine whether better silica actually lowers downstream purification cost, improves yield or gives a differentiated product. Do not infer semiconductor suitability from SiO_2 assay alone.

PRIORITY 3 - QUALIFY THE WHOLE SUPPLY CHAIN

Secure reliable suppliers for wafers, gases, chemicals, masks, slurries, packaging and test. Establish quality agreements, traceability, change control and contingency supply before scaling.

PRIORITY 4 - BUILD FOUNDRY CAPABILITY IN STAGES

Develop process engineering, metrology and cleanroom expertise through collaboration and pilot work. Only consider owning an integrated fab once process integration, customer demand, capital, utilities, permitting and equipment support are credible.

The practical first-year work package

- Complete the SilicaFlux product specification and freeze a small first-chip scope.
- Resolve RTL format, synthesis path and verification; demonstrate key compute blocks on FPGA.
- Shortlist foundry processes and obtain PDK access, MPW options, quotations and engineering support.
- Build a quartz sampling and assay programme with a competent mineral laboratory.
- Run small-scale beneficiation and reduction trials with measured mass and impurity balances.
- Create a supplier qualification register for wafers, process chemicals, gases, packaging and test.
- Prepare a stage-gated investment model that distinguishes design costs, prototype tape-out, materials pilot and full fab capital.

References and technical starting points

The sources below support the process overview and identify useful standards or public documentation. They are starting points for engineering due diligence, not substitutes for foundry process documentation, supplier specifications, legal advice or a site-specific engineering design.

[1] OSHA - Semiconductor: Substrate Manufacture.
https://www.osha.gov/semiconductors/silicon/substrate-manufacture
Overview of polysilicon production from quartz, trichlorosilane purification, Czochralski crystal growth, ingot machining and wafer preparation.

[2] Semiconductor Industry Association - Stage 3: Front-end Manufacturing.
https://www.semiconductors.org/semiconductors-101/how-are-semiconductors-made/stage-3-front-end-manufacturing/
Accessible summary of oxidation, lithography, doping, deposition, metallisation, etching, CMP and metrology.

[3] SEMI - Standards FAQ: Ultrapure Water.
https://www.semi.org/en/products-services/standards/standardsfaq
Identifies SEMI guidance for ultrapure-water systems and filtration in semiconductor processing.

[4] SkyWater / Google - SKY130 Open Source PDK documentation.
https://skywater-pdk.readthedocs.io/en/main/
Public PDK documentation; current documentation describes the release as an experimental preview and cautions that it is not intended for production settings.

[5] UK Department for Science, Innovation and Technology - National Semiconductor Strategy (2023).
https://www.gov.uk/government/publications/national-semiconductor-strategy
UK policy context for semiconductor design, manufacturing, supply-chain resilience and the stated cost scale of state-of-the-art foundries.

[6] OSHA - Silicon Device Manufacturing.
https://www.osha.gov/semiconductors/silicon
Overview of substrate manufacture, device fabrication, metallisation, wafer sort, dicing, die attach, bonding and packaging.

[7] NIST - Semiconductor Manufacturing: Background Information (NISTIR 7916).
https://nvlpubs.nist.gov/nistpubs/ir/2013/NIST.IR.7916.pdf
Technical background on deposition, etching, lithography, electrical modification and metrology in semiconductor fabrication.

Planning assumptions and limitations

This report is a conceptual manufacturing roadmap. It does not specify process recipes, chemical concentrations, furnace designs, cleanroom classifications, equipment models, plant capacities, yield guarantees or capital costs for a particular site. These require a selected device process, foundry or technology partner, customer specifications, site utility study, environmental permits, process-hazard analysis and detailed engineering. Semiconductor processes are proprietary and tool-specific; actual operating conditions must be taken from qualified vendor and foundry documentation.

The report is focused on conventional silicon CMOS. Compound semiconductors, power devices, MEMS, photonics and other device classes require different materials, substrates and process flows.